Olist E-Commerce Analytics: SQL to Power BI

End-to-end analytics project | PostgreSQL, Power BI, DAX | Brazilian marketplace data (currency: R$)

1. Executive Summary

This project turns the public Olist e-commerce dataset (nine relational tables, 112,650 order items, 98,666 orders and 3,095 sellers) into a five-page Power BI dashboard. It covers the complete analytics pipeline: loading raw CSV files into PostgreSQL, enforcing a relational model with primary and foreign keys, shaping the data into BI-ready SQL views, building a DAX measure library, and presenting sales, product, delivery and payment performance for decision-makers.

Headline results

- The marketplace generated **R$13.59M** in item revenue plus **R$2.25M** in freight, at an average order value of **R$137.75**.
- Delivery is largely reliable: **6.8%** of delivered orders (6,534 of 96,478) arrived after the estimated date, so 93.2% were on time.
- Credit cards dominate payments (**75.4%** of orders), and about two thirds of credit card orders are paid in installments.
- Freight adds about R$16.6 for every R$100 of product value, a meaningful cost burden on customers.

2. Business Problem and Objectives

An online marketplace needs a single, trustworthy view of how the business is performing. This project was built to answer four groups of questions:

- **Sales:** How are revenue and order volume trending, and how does this period compare with the same period last year?
- **Products and satisfaction:** Which categories drive revenue and orders, and how do customers rate them?
- **Delivery:** How reliable is delivery, where are the delays, and how does the shipping route (same state versus cross state) affect delivery time, freight cost and lateness?
- **Sellers and payments:** How do customers pay, and how is revenue spread across sellers and regions?

The objective was one governed data model and one dashboard that moves from executive KPIs to operational drill-downs.

3. Data and Data Model

The source is the public Olist dataset, supplied as nine CSV files. Each was loaded into its own PostgreSQL table with an explicit primary key and foreign keys that mirror the real business relationships.

Table	Grain (one row per)	Role
customers_data	customer_id	Customer city, state and unique customer id
orders_data	order	Status and purchase, approval, carrier, delivery and estimated dates
order_items_data	order item (order_id + order_item_id)	Price, freight, product and seller
order_payments_data	payment (order_id + payment_sequential)	Payment type, installments, value
order_reviews_data	review	Review score and comments
products_data	product	Category and physical dimensions
sellers_data	seller	Seller city and state
product_category_name_translation	category	Portuguese to English category names
geolocations_data	zip prefix	Latitude and longitude reference

Foreign keys: orders reference customers; order items reference orders, products and sellers; payments reference orders; reviews reference orders. These constraints guarantee that no item, payment or review can exist without a valid parent order.

4. Data Engineering in PostgreSQL

4.1 Loading the data: problems found and how they were solved

Importing the files was not a single click. Each problem below had a specific cause and fix, and together they show why raw data should be profiled before it is loaded.

Problem	Cause	Fix
Date values rejected or misread	Source dates use month/day/year; PostgreSQL expected another order	Set datestyle to MDY at database level so every session and import wizard inherits it
Import rejected for the category translation file	The file has a .csv extension but is tab-separated	Import with a tab delimiter instead of a comma
Encoding error on the reviews file	Accented Portuguese characters saved in a Windows code page, not UTF-8	Set client encoding to match the file before importing
Duplicate key errors on items and payments	One order can contain several items and several payment rows, so order_id is not unique	Use composite primary keys: (order_id, order_item_id) and (order_id, payment_sequential)
Rows doubled after a repeated import	The same file was loaded twice	Truncate before reloading; the primary key now acts as a safety net against accidental duplicates
Server-side copy could not find the file	Server-side COPY reads files as the database service account, not the logged-in user	Use the client-side psql copy command with the exact file path

4.2 BI-ready SQL views

Rather than loading nine normalised tables into Power BI and rebuilding joins there, the business logic was placed in four SQL views. This keeps transformations in one auditable place and lets the dashboard import a single flat fact table.

- **bi_dim_product:** attaches the English category name (falling back to the Portuguese name when no translation exists) and computes product volume from dimensions.
- **bi_dim_latest_review:** keeps only the most recent review per order using DISTINCT ON, because some orders have more than one review.
- **bi_payments_order:** collapses payments to one row per order, holding total payment value, maximum installments and the primary payment type (the type that carried the largest amount).
- **bi_fact_order:** one row per order with purchase date and month, delivery days (purchase to delivery, delivered orders only), an is_late flag, and customer city

and state. An order is late when it is delivered and the actual delivery date is after the estimated date.

- **bi_fact_sales:** the item-grain fact table that joins items to the order, product, latest review, order-level payments and seller attributes. This is the one table imported into Power BI.

4.3 Controlling grain: the most important design decision

A fact table must have a clear grain. Here the grain is one row per order item. Payments and reviews, however, exist at order level, and one order can have several payment rows (this affects 2,961 of 99,440 orders in the payments file).

Joining the raw payments table directly to order items multiplied rows: 112,650 items became 117,604 rows, with 3,284 duplicated item keys, and the summed payment value was overstated by roughly 27%. The final design therefore collapses payments and reviews to **one row per order before joining**, so the fact table keeps exactly one row per order item.

A related point remains: order-level attributes such as payment value and installments are repeated on every item row of the same order. Measures that aggregate them must count each order once (see Appendix B). Item-level attributes such as price and freight can be summed normally.

5. Analytics Layer: DAX Measures

Because the fact table is item grain, order-based measures use DISTINCTCOUNT on order_id so that multi-item orders are never counted more than once.

Group	Measures
Sales	Total Revenue, Total Order, Total Customer, AOV, Item Sold, Item per Order, Total Freight Value
Delivery	Order Delivered, Late Orders, Late Delivery %, On Time %, Avg Delivery Days
Time intelligence	Revenue LY, Revenue YTD, Rolling 30D Revenue, YoY %
Sellers and payments	Total Sellers, Revenue per Seller, Avg Freight per Item, Total Payment, Payment Installments
Reviews	Positive Reviews %, Negative Review %, Review Bucket (calculated column)

Calculated columns and tables	Route Type (same state or cross state), Country, Dim_Date calendar table

Key definitions

- **Late Delivery %** = late delivered orders divided by delivered orders. Cancelled, unavailable and in-transit orders are excluded from the denominator, so the rate is not diluted.
- **Route Type** compares seller state with customer state, classifying each item as Same state, Cross state or Unknown.
- **Dim_Date** is a dedicated calendar table generated from the first to last purchase date. It powers year-over-year, year-to-date and rolling-window measures.

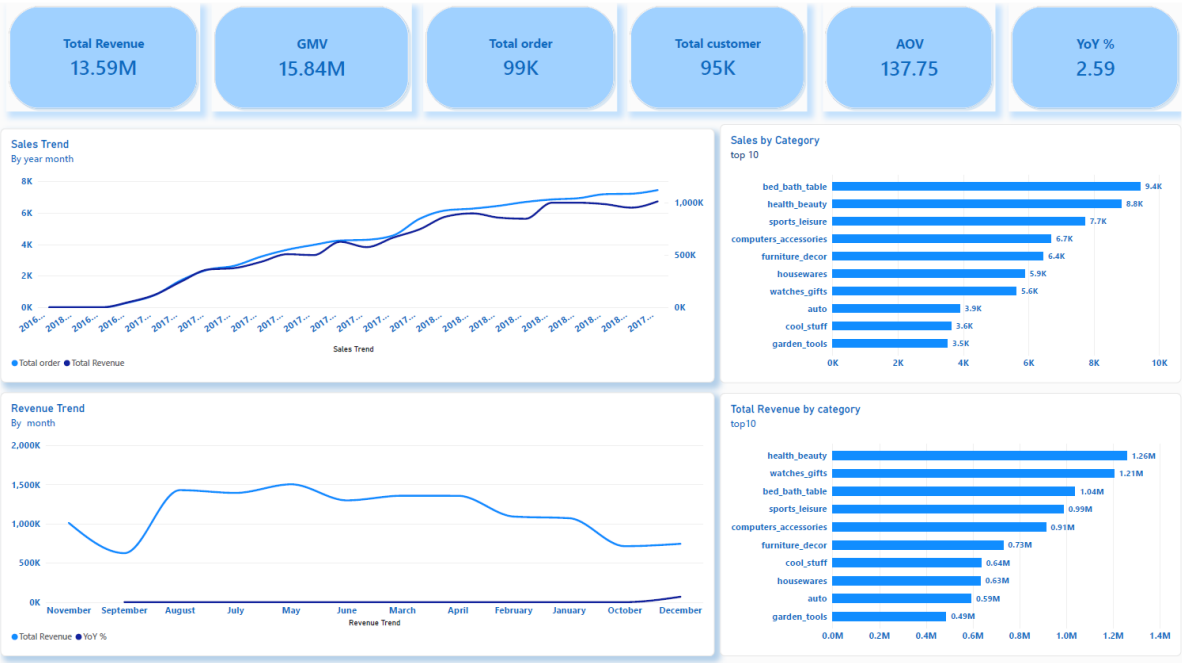
6. Dashboard Walkthrough

The report has six pages that move from a high-level summary to specific operational questions.

Page: overview (Excutive dashboard).

Purpose: One screen business health check.

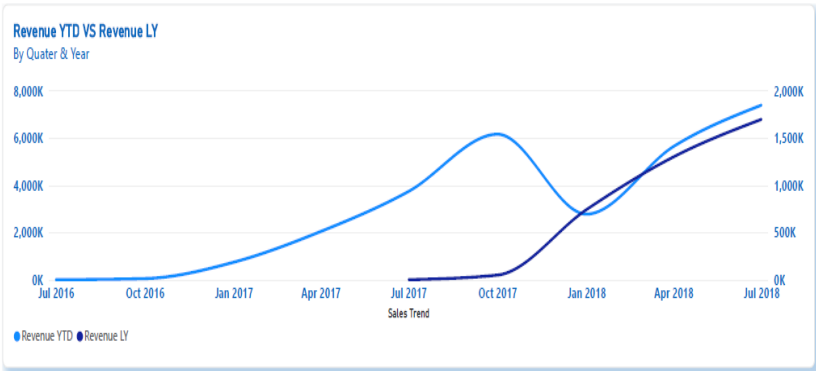
Visual: KPI cards (Total revenue, Total order, total customers, AOV, GMV, YOY%) revenue trend with yoy % , order and revenue by month, sales by category.

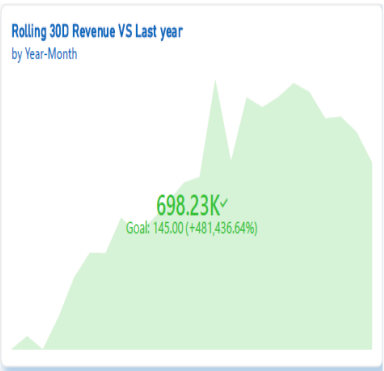


Page: Sales Trend

Purpose: Growth & seasonality.

Visual: Revenue YTD (year to date) Vs last year, rolling 30 days revenue vs last year matrix of month by category and revenue, order and AOV.



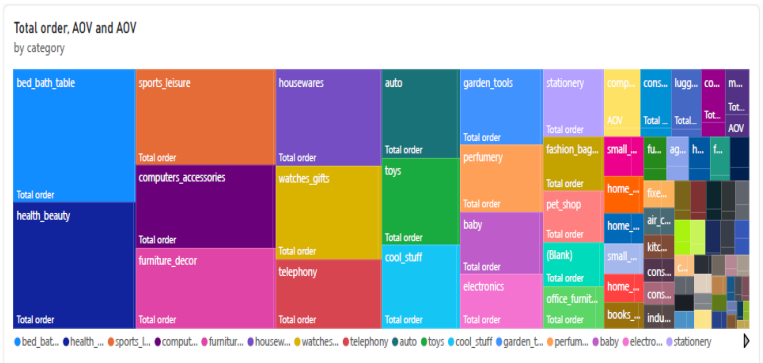


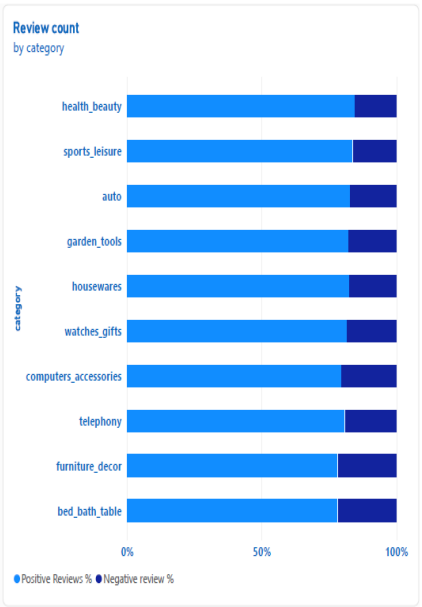
Year-Month	2016-09			2016-10			2016-12			2017-01			2017-02			2017-03			2017-04		
category	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV
				65.89	2	32.95				1,702.64	11	154.79	9,034.10	52	173.73	6,714.17	57	117.79	8,693.09	63	137.99
agro_industry_and_commerce										65.97	2	32.99	224.84	7	32.12	81.99	2	41.00			
air_conditioning				1,707.09	5	341.42				663.70	3	221.23	3,095.30	12	257.94	4,103.81	16	256.49	2,582.23	14	184.45
art																223.25	5	44.65	119.90	1	119.90
arts_and_craftmanship																					
audio				156.99	2	78.50							163.80	5	32.76	1,134.77	13	87.29	942.20	9	104.69
auto				1,833.25	11	166.66				5,218.53	30	173.95	13,162.40	77	170.94	14,482.07	76	190.55	15,548.17	81	191.95
baby				1,630.16	11	148.20				6,397.87	32	199.93	3,048.48	33	92.38	3,724.08	43	86.61	3,980.45	48	82.93
bed_bath_table				478.99	5	95.80				3,960.16	45	88.00	16,282.73	146	111.53	25,773.02	249	103.51	24,347.69	234	104.05
books_general_interest				119.50	1	119.50				234.89	2	117.45	772.71	8	96.59	2,648.96	20	132.45	923.38	12	76.95
books_imported													99.00	1	99.00				19.99	1	19.99
books_technical				267.00	1	267.00							159.00	1	159.00	75.00	2	37.50	75.00	1	75.00
cds_dvds_musicals																					
christmas_supplies																			69.00	1	69.00
cine_photo													59.00	1	59.00				118.00	2	59.00
computers										2,199.99	1	2,199.99	3,549.00	1	3,549.00						
computers_accessories				1,399.32	18	77.74				3,924.14	24	163.51	11,972.59	97	123.43	28,624.60	150	190.83	20,691.06	120	172.43
consoles_games				3,882.26	10	388.23				5,954.20	21	283.53	4,294.77	18	238.60	10,768.84	48	224.35	3,772.88	15	251.53
construction_tools_construction										310.00	1	310.00	336.79	3	112.26	773.40	4	193.35			
construction_tools_lights																					
construction_tools_safety													379.00	1	379.00	380.00	2	190.00			
Total	267.36	3	89.12	49,507.66	308	160.74	10.90	1	10.90	120,312.87	789	152.49	247,303.02	1733	142.70	374,344.30	2641	141.74	359,927.23	2391	150.53

Page: category and product.

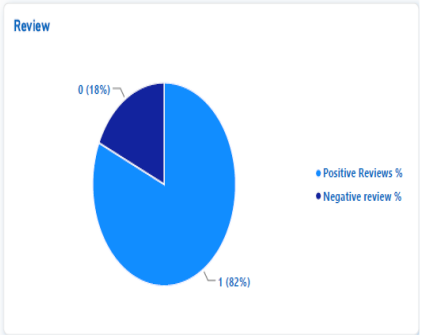
Purpose: product mix and customer sentiments.

Visual: positive verses negative review split, review sentiment by category (100% stacked) treemap of category by order and AOV, month by category matrix.





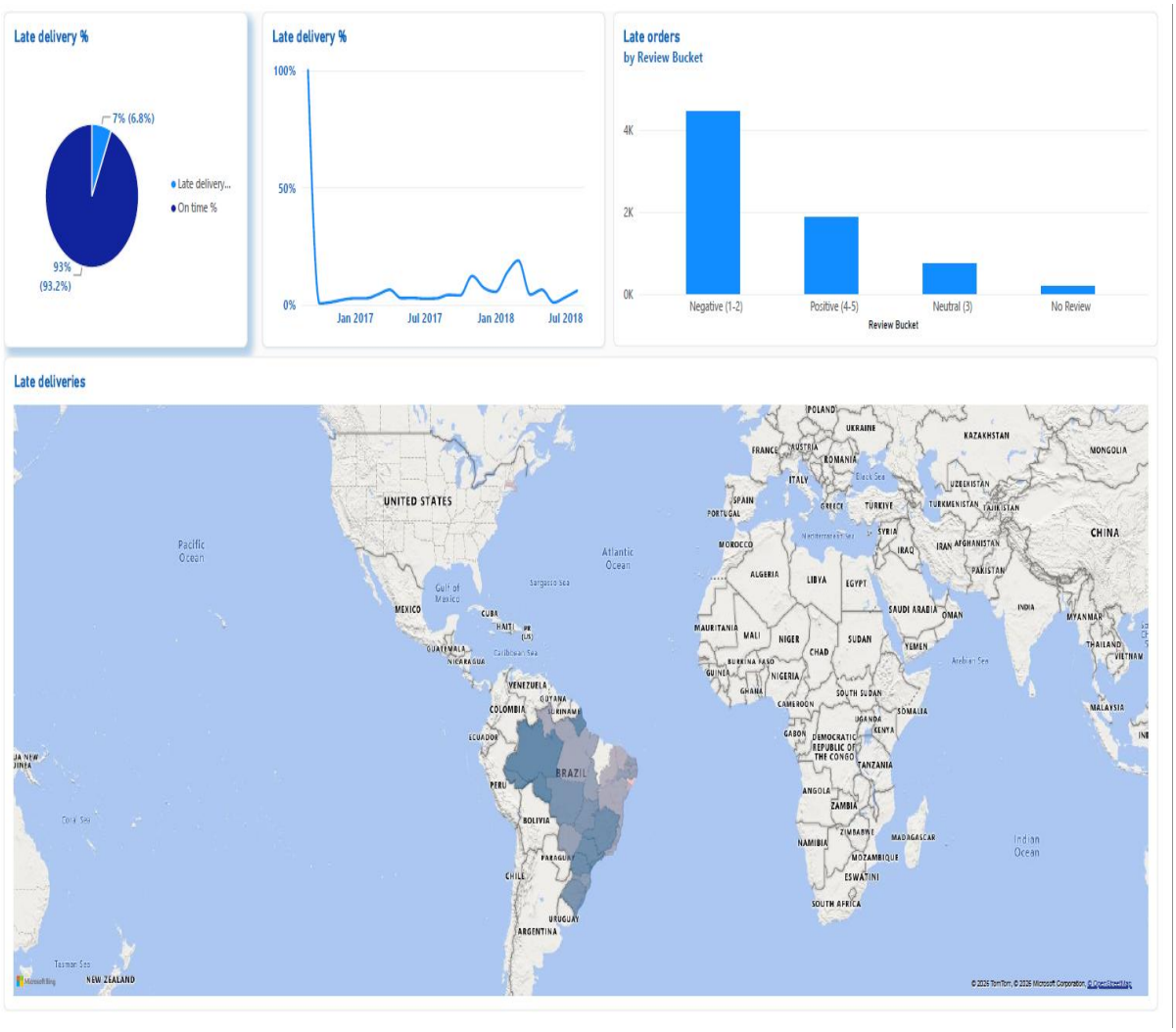
Year	2016			2017			2018			Total		
category	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV	Total Revenue	Total order	AOV
	65.89	2	32.95	106,047.31	822	129.01	73,422.08	627	117.10	179,535.28	1451	123.7
agro_industry_and_commerce				29,179.20	48	607.90	43,351.27	134	323.52	72,530.47	182	398.5
air_conditioning	1,707.09	5	341.42	28,253.01	114	247.83	25,064.86	134	187.05	55,024.96	253	217.4
art				9,291.79	36	258.11	14,910.85	166	89.82	24,202.64	202	119.8
arts_and_craftmanship				151.89	2	75.95	1,662.12	21	79.15	1,814.01	23	78.8
audio	156.99	2	78.50	17,570.57	161	109.13	32,960.94	187	176.26	50,688.50	350	144.8
auto	1,833.25	11	166.66	243,255.71	1429	170.23	347,631.15	2457	141.49	592,720.11	3897	152.1
baby	1,630.16	11	148.20	153,334.03	1207	127.04	256,800.70	1667	154.05	411,764.89	2885	142.7
bed_bath_table	478.99	5	95.80	498,440.43	4503	110.69	538,069.26	4909	109.61	1,036,988.68	9417	110.1
books_general_interest	119.50	1	119.50	19,630.49	221	88.83	27,106.89	290	93.47	46,856.88	512	91.5
books_imported				1,123.99	13	86.46	3,515.86	40	87.90	4,639.85	53	87.5
books_technical	267.00	1	267.00	5,762.22	43	134.01	13,066.84	216	60.49	19,096.06	260	73.4
cds_dvds_musicals				665.00	11	60.45	65.00	1	65.00	730.00	12	60.8
christmas_supplies				2,854.92	50	57.10	5,945.90	78	76.23	8,800.82	128	68.7
cine_photo				695.00	8	86.88	6,238.46	57	109.45	6,933.46	65	106.6
computers				159,230.79	123	1,294.56	63,732.34	58	1,098.83	222,963.13	181	1,231.8
computers_accessories	1,399.32	18	77.74	405,078.69	2618	154.73	505,476.31	4053	124.72	911,954.32	6689	136.3
consoles_games	3,882.26	10	388.23	88,394.45	665	132.92	65,188.51	387	168.45	157,465.22	1062	148.2
construction_tools_construction				20,315.28	105	193.48	124,362.31	643	193.41	144,677.59	748	193.4
Total	49,785.92	312	159.57	6,155,806.98	44579	138.09	7,386,050.80	53775	137.35	13,591,643.70	98666	137.7



Page: Delivery and logistics.

Purpose: Delivery reliability.

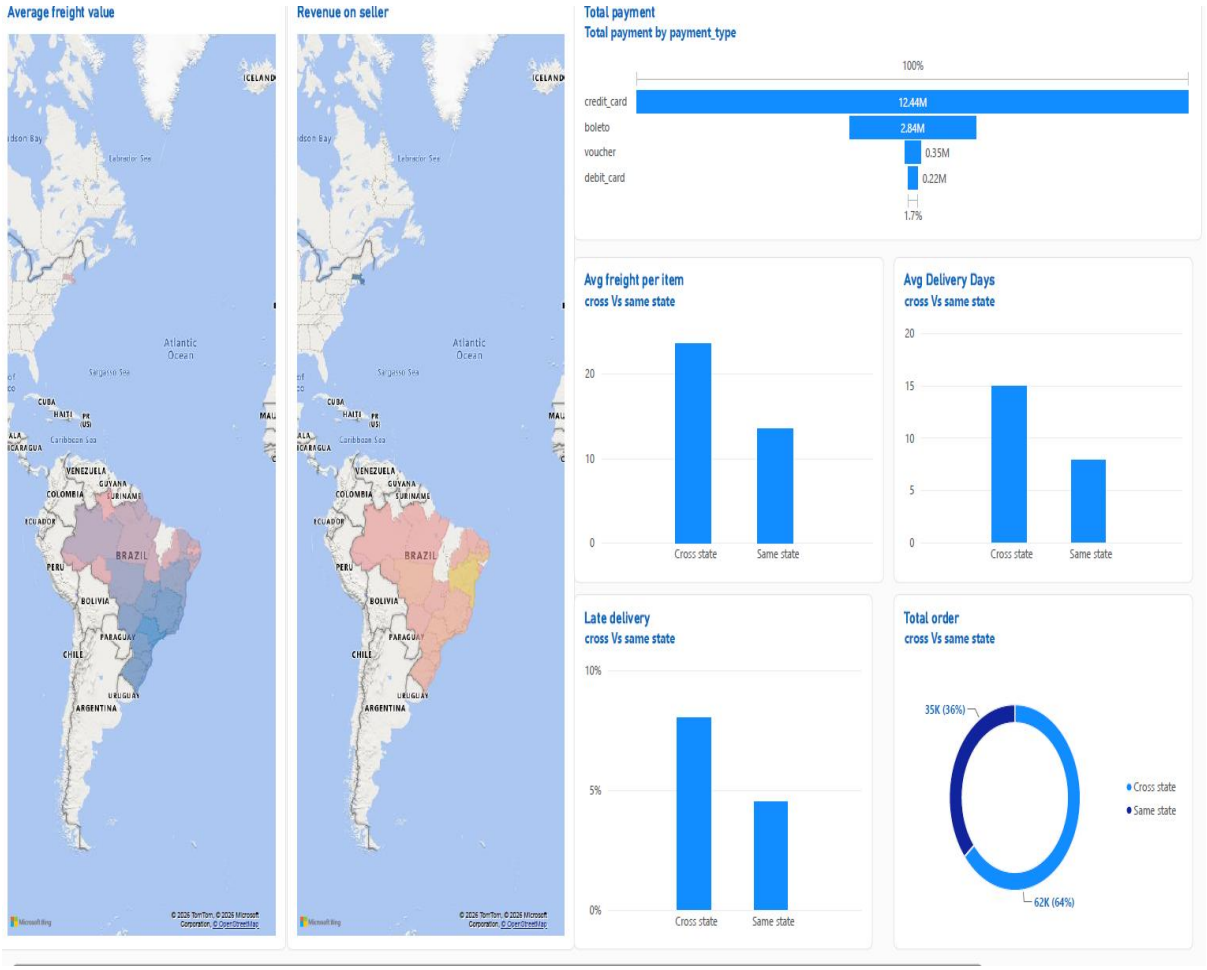
Visual: field map of late delivery % by customer state, monthly late delivery % by average delivery days, late versus on time delivery, late order by delivery bucket.



Page: sellers and payment.

Purpose: coast root and payment behaviour .

Visual: same state versus cross state camparison (late %, delivery days, freight per item, share of orders), seller map(revenue per seller), freight map by customer state, payment funnel by type, installments and values.



Page: Insight and Recommendations.

Purpose: Sales and basket, delivery, payment and freight, recommendation, footnote.

Visual: text data (not visual) .

INSIGHT & RECOMMENDATIONS :

SALES & BASKET

- R$13.59M item revenue from 98,666 orders and 112,650 items
- Average order value is R$137.75
- 90% of orders contain a single item (1.14 items per order), so growth depends on winning more orders
- 3,095 active sellers earn about R$4,391 each on average

DELIVERY

- 6.8% of delivered orders were late (6,534 of 96,478)
- 93.2% arrived on or before the estimated date
- Highest late-delivery state: [Al] at [21.4]% ← see map on Delivery page
- Lowest late-delivery state: [Ro] at [2.9]%
- Cross-state orders vs same-state: [15] vs [8] days ← see Sellers & Payments page

PAYMENTS & FREIGHT

- Credit card: 75.4% of orders, 78.3% of payment value
- Boleto 19.9%, voucher 3.2%, debit card 1.5% of orders
- 66.9% of credit card orders are paid in installments (avg 3.5)
- Freight is R$2.25M, equal to 16.6% of item revenue

RECOMMENDATIONS

- Manage freight cost: at 16.6% of item revenue it is a visible burden. Find the states and routes with the highest freight per item and test carrier negotiation or free-shipping thresholds there.
- Fix delivery by region: focus carrier and seller follow-up on the states with the highest late rates instead of acting on the 6.8% average.
- Bring sellers closer to customers: if same-state orders deliver faster and cost less freight, recruit sellers in under-served states.
- Protect installment options: most credit card orders use installments, so any change to terms affects the majority of revenue.
- Track delays against reviews: monitor whether late orders pull review scores down, and alert on rising late rates.

FOOTNOTE

Revenue = sum of item price across all order statuses. Late = delivered after the estimated date. Order counts are distinct orders.

7. Key Findings

All figures below were calculated from the order items and payments source data and match the dashboard definitions.

7.1 Scale and value

Metric	Value
Orders with items	98,666
Order items sold	112,650
Items per order	1.14 (9.9% of orders contain more than one item)
Item revenue (sum of price)	R$13.59M
Freight charged	R$2.25M (16.6% of item revenue)
Item value plus freight	R$15.84M
Average order value	R$137.75
Active sellers	3,095
Distinct products	32,951
Revenue per seller	about R$4,391

Most orders are single-item purchases, so growth depends mainly on winning more orders rather than increasing basket size.

7.2 Delivery performance

6.8% of delivered orders were late (6,534 of 96,478), and **93.2%** arrived on or before the estimated date. At marketplace level this is a healthy result, which means the useful analysis lies in where delays concentrate. The Delivery and Logistics page breaks lateness down by state and by month, and the Sellers and Payments page compares same-state and cross-state routes on lateness, delivery time and freight.

7.3 Payment behaviour

Payment type	Share of orders	Share of payment value
Credit card	75.4%	78.3%
Boleto (bank slip)	19.9%	17.9%

Voucher	3.2%	2.4%
Debit card	1.5%	1.4%

Share of orders uses each order's primary payment type (the type carrying the largest amount). Credit card orders average 3.5 installments, and **66.9%** of them are paid in more than one installment. Installment financing is therefore central to how customers buy.

7.4 Freight burden

Freight equals 16.6% of product revenue, so a customer pays roughly R$16.6 in shipping for every R$100 of goods. Because freight is charged per item and varies by distance, it is a lever worth tracking by region and by route type, which is why the dashboard includes freight per item by state and by route.

8. Analytical Decisions and Data Quality Notes

- **Latest review only:** an order with several reviews is represented by its most recent one, so review metrics are not inflated by repeat reviews.
- **One payment row per order in the fact table:** the primary payment type is chosen by the largest amount, a documented and repeatable rule.
- **Delivered-only delivery metrics:** delivery days and the late flag are calculated only for delivered orders with valid dates.
- **Order counts use DISTINCTCOUNT:** this prevents multi-item orders from being counted once per item.
- **Honest interpretation:** route type and lateness show association, not proof of cause. Distance, carrier and seller quality can all contribute.

9. Recommendations

1. **Manage freight cost actively.** At 16.6% of item revenue, freight is a visible cost for customers. Use the freight-by-state and route-type views to find the regions and routes where shipping is most expensive, then test carrier negotiation or free-shipping thresholds there.
2. **Target delivery improvement by region.** Use the late delivery map to focus carrier and seller follow-up on the states with the highest late rates, rather than acting on the 6.8% average.
3. **Keep installment options.** With most credit card orders paid in installments, any change to installment terms would affect the majority of revenue.

4. **Bring sellers closer to customers.** The same-state versus cross-state comparison shows whether local fulfilment shortens delivery and lowers freight. If it does, recruiting sellers in under-served states is a practical lever.
5. **Monitor delays alongside satisfaction.** The late-orders-by-review view lets the business track whether late deliveries are pulling review scores down.

10. Limitations and Next Steps

- Total Revenue sums item price across all order statuses, so it includes cancelled and unavailable orders. A delivered-only revenue measure would be stricter.
- The data is historical (2016 to 2018), and the earliest and latest months are typically partial, so year-over-year comparisons are only reliable for periods that are complete in both years.
- Delay analysis does not yet separate seller handling time from carrier transit time. Adding the carrier handover date would split the delivery journey into those two stages and point to who should fix the delay.
- Next steps: publish to the Power BI service with scheduled refresh, add a delivered-only revenue view, and add seller scorecards with a minimum-order threshold.

Appendix A: Key SQL

Order-level payment view

```
CREATE OR REPLACE VIEW bi_payments_order AS
SELECT
  op.order_id,
  SUM(op.payment_value)       AS order_payment_value,
  MAX(op.payment_installments) AS order_payment_installments,
  (ARRAY_AGG(op.payment_type ORDER BY op.payment_value DESC NULLS LAST))[1]
AS order_payment_type
FROM order_payments_data AS op
GROUP BY op.order_id;
```

Delivery days and late flag (inside bi_fact_order)

```
CASE WHEN o.order_status = 'delivered'
    AND o.order_purchase_timestamp IS NOT NULL
```

```
    AND o.order_delivered_customer_date IS NOT NULL
   THEN o.order_delivered_customer_date::date - o.order_purchase_timestamp::date
END AS delivery_days,
CASE WHEN o.order_status = 'delivered'
    AND o.order_delivered_customer_date IS NOT NULL
    AND o.order_estimated_delivery_date IS NOT NULL
    AND o.order_delivered_customer_date::date >
o.order_estimated_delivery_date::date
   THEN 1 ELSE 0
END AS is_late
```

Payment join in the item-grain fact (inside bi_fact_sales)

```
 p.order_payment_type       AS payment_type,
 p.order_payment_installments AS payment_installments,
 p.order_payment_value      AS payment_value,
 ...
LEFT JOIN bi_payments_order AS p ON p.order_id = oi.order_id
```

Appendix B: Core DAX

```
Total Revenue = SUM(bi_fact_sales[price])
Total Order   = DISTINCTCOUNT(bi_fact_sales[order_id])
AOV           = DIVIDE([Total Revenue], [Total Order])

Order Delivered =
CALCULATE([Total Order], bi_fact_sales[order_status] = "delivered")

Late Orders =
CALCULATE(
   [Total Order],
   bi_fact_sales[order_status] = "delivered",
```

```
    bi_fact_sales[is_late] = 1
)

Late Delivery % = DIVIDE([Late Orders], [Order Delivered])
On Time %      = DIVIDE([Order Delivered] - [Late Orders], [Order Delivered])

-- one value per order, not per item
Avg Delivery Days =
AVERAGEX(VALUES(bi_fact_sales[order_id]),
        CALCULATE(MAX(bi_fact_sales[delivery_days])))
Total Payment =
SUMX(VALUES(bi_fact_sales[order_id]),
    CALCULATE(MAX(bi_fact_sales[payment_value])))
Revenue LY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))
YoY %     = DIVIDE([Total Revenue] - [Revenue LY], [Revenue LY])

Route Type =
IF(ISBLANK(bi_fact_sales[seller_state]) || ISBLANK(bi_fact_sales[customer_state]),
  "Unknown",
  IF(bi_fact_sales[seller_state] = bi_fact_sales[customer_state],
     "Same state", "Cross state"))
```